

Constrained Allocation Reinforcement Learning with Hard Simplex Feasibility and Adaptive Path-Risk Control

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Sequential allocation problems require an agent to repeatedly distribute a finite, divisible resource among competing entities. Their action space is naturally a probability simplex, but practical requirements introduce two fundamentally different forms of constraint. Hard allocation constraints restrict every realised action, whereas path-dependent constraints regulate cumulative behaviour over a trajectory. This report develops a hybrid reinforcement-learning (RL) formulation that treats these requirements separately. Constrained Allocation Optimization with Simplex Decomposition (CAOSD) parameterises actions satisfying two possibly overlapping group-exposure constraints by construction. Reward Constrained Policy Optimization (RCPO) then adapts a Lagrange multiplier for a soft trajectory cost without altering the hard-feasible action set. The framework is instantiated in portfolio allocation, where assets are the allocation entities, minimum group exposures are hard constraints, and benchmark-relative drawdown is the path cost. To study credit assignment within the decomposed policy, we compare standalone branch returns with an open-loop counterfactual that replaces one realised branch by a neutral allocation while holding the remaining branches fixed. Under matched 60,000-update interaction budgets and evaluation on 20 unseen synthetic market continuations, counterfactual-reward Dirichlet RCPO attains the highest observed mean terminal relative wealth, 8.18%, but improves on its standalone-credit control by only 0.12 percentage points and increases turnover. Counterfactual-cost Gaussian RCPO produces the clearest drawdown improvement over its matched control, whereas joint counterfactual reward and cost is not consistently beneficial. The structural constraints are satisfied for every evaluated action; reliable out-of-sample satisfaction of the path-risk target is not achieved. The results therefore support the general separation of hard action feasibility from adaptive cumulative-risk control, while empirical claims remain limited to a single-seed portfolio study in synthetic markets.

1 INTRODUCTION AND CONTRIBUTIONS

Allocation is a recurring decision primitive in finance, computing, logistics, communication networks, and energy systems. At each time step, an agent distributes a limited resource among a set of entities, observes the consequences, and reallocates in response to a changing environment. Examples include dividing capital among assets, assigning workload to servers, distributing power across devices, and allocating capacity among users. Reinforcement learning (RL) is attractive in these settings because an allocation can influence both immediate utility and the state in which future allocations are chosen. The difficulty is that feasible resource use is rarely described by the simplex constraint alone.

Two classes of constraint arise. First, *hard allocation constraints* prescribe which actions may be executed at an individual decision. Group quotas, capacity shares, diversification rules, and minimum service levels belong to this class. Violating such a rule once may make the action operationally invalid, even when the policy is feasible on average. Second, *soft cumulative constraints* regulate behaviour over time. Long-run energy use, average congestion, cumulative risk, and drawdown-related costs depend on the realised trajectory and are naturally represented through cost returns. Conflating the two classes is problematic: a finite expected penalty does not provide a policy-support guarantee, while an action-space construction cannot by itself control a path statistic.

The portfolio application makes the distinction concrete. Long-only weights and group-exposure requirements define a static feasible region that every rebalance must respect. Drawdown depends on the complete wealth path and on previous peaks, so it cannot be characterised by the current weight vector alone. Portfolio RL also introduces adjustment costs and generalisation concerns: rebalancing changes realised return, and strong performance on one market path need not transfer to another [5]. These features make portfolio allocation a useful instance of the broader constrained allocation problem rather than a separate mathematical category.

1.1 RELATED WORK AND POSITIONING

Constrained Markov decision processes (CMDPs) augment the reward objective with expected cumulative cost limits. Constrained Policy Optimization (CPO) uses trust-region arguments to obtain near-constraint satisfaction during policy updates [7], whereas Reward Constrained Policy Optimization (RCPO) uses a multi-timescale Lagrangian mechanism to adapt a penalty multiplier [2]. These and related safe-RL methods are well suited to cumulative behavioural requirements, but an expectation constraint does not generally imply that every sampled action is feasible [4].

Action-constrained RL instead treats admissibility as a property of the action set. Projection and safety-layer methods can enforce zero violation, although projection may couple correction and policy gradients or introduce optimisation overhead [8]. Constrained allocation is a structured action-constrained setting because non-negative resource shares sum to one and group restrictions cut a convex polytope from that simplex. CAOSD exploits the special geometry of two allocation constraints by combining simplex-valued sub-actions through a weighted Minkowski sum [1]. More recent work, such as Polytope Action Space Policy Optimization (PASPO), uses autoregressive allocation and linear programming to handle more general convex polytopes [10]. The present work does not implement that broader capability: its hard-feasibility mechanism is deliberately restricted to two, possibly overlapping, group-exposure constraints.

The proposed architecture combines this specialised hard-action map with a cumulative-cost learner. Proximal Policy Optimization (PPO) [3] optimises a stochastic policy on the decomposed surrogate space; CAOSD deterministically maps its branch actions to the feasible allocation space; and RCPO adjusts behaviour within that space using the realised path cost. Thus, structural feasibility is not delegated to a stochastic penalty, and cumulative risk is not incorrectly treated as a static geometric constraint.

The work makes three contributions:

1. It formulates sequential constrained allocation with separate requirements for hard, per-decision feasibility and soft, cumulative trajectory cost, and maps portfolio allocation into this general model.
2. It combines a two-group CAOSD policy with RCPO so that the policy remains allocation-feasible for every action while learning from an observable path-dependent risk signal.
3. It evaluates standalone and open-loop counterfactual credit assignment for the dependent branches of the decomposed policy under matched portfolio experiments.

The intended contribution is methodological. The report does not claim support for arbitrary allocation polytopes, guaranteed drawdown control, or profitability in real markets.

2 PROBLEM FORMULATION

2.1 SEQUENTIAL ALLOCATION AND INFORMATION STRUCTURE

Consider a finite-horizon constrained Markov decision process (CMDP) [7],

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma, C), \quad C = (C_1, \dots, C_Q), \quad (1)$$

with initial distribution ρ_0 , horizon H , and cost budgets d_q . Here P is the transition kernel, R the reward function, $\gamma \in (0, 1]$ the discount factor, and C_q integrable transition costs. To separate exogenous conditions from decision memory, write

$$s_t = (x_t, a_{t-1}, \zeta_t, t), \quad (2)$$

where x_t contains sufficient market or system variables and ζ_t contains endogenous path statistics. In the simulator, the exogenous transition is determined by latent regime and momentum dynamics, or by market identity and index when a fixed tape is replayed. Initial allocation and memory belong to ρ_0 .

The actor observes $o_t = \mathcal{O}(s_t)$, not the full state: trailing returns, rolling mean and volatility, previous target weights and turnover, allocation and drawdown diagnostics, and elapsed time. Its policy is non-anticipative with respect to available information $\mathcal{F}_t$; future returns and latent regimes are excluded. Previous allocation is necessary because identical market information and new actions can incur different costs under different prior allocations. Wealth and peak statistics similarly distinguish histories with different drawdowns. This augmentation makes endogenous costs Markovian, but a finite return window need not identify the latent market state; the implemented actor remains an observation-conditioned approximation to partially observed control.

2.2 HARD FEASIBILITY AND CUMULATIVE CONSTRAINTS

For N allocation entities $\mathcal{E} = \{1, \dots, N\}$, the nominal action space is

$$\mathcal{A} = \Delta^N := \{\mathbf{a} \in \mathbb{R}^N : a_i \geq 0 \ (i \in \mathcal{E}), \ \mathbf{1}^\top \mathbf{a} = 1\}. \quad (3)$$

The superscript counts coordinates; the affine dimension is $N - 1$. For groups $V_k \subseteq \mathcal{E}$ and limits $0 \leq \ell_k \leq u_k \leq 1$, assume the hard-feasible set is non-empty:

$$\mathcal{A}_h = \left\{ \mathbf{a} \in \Delta^N : \ell_k \leq \sum_{i \in V_k} a_i \leq u_k, \ k = 1, \dots, K \right\} \neq \emptyset. \quad (4)$$

An upper bound is equivalent to $\sum_{i \in \mathcal{E} \setminus V_k} a_i \geq 1 - u_k$. The implemented CAOSD map supports only two lower-bound group constraints, not arbitrary polytopes. Define Π_h as the non-anticipative policies satisfying

$$\pi_t(\mathcal{A}_h | \mathcal{F}_t) = 1 \quad \text{almost surely}, \quad t = 0, \dots, H - 1. \quad (5)$$

This support condition guarantees feasibility of every executed action. An unconstrained continuous density assigns zero mass to the unit-sum hyperplane, and a finite penalty does not enforce this support. Softmax and Dirichlet policies satisfy the simplex but not necessarily group exposures [8, 1].

Let $r_t = R(s_t, \mathbf{a}_t, s_{t+1})$ and $c_{q,t+1} = C_q(s_t, \mathbf{a}_t, s_{t+1})$. Reward may include adjustment costs before a nonlinear utility transformation. Define

$$J_R(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{H-1} \gamma^t r_t \right], \quad (6)$$

$$J_{C_q}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{H-1} \gamma^t c_{q,t+1} \right]. \quad (7)$$

The general constrained-allocation problem is

$$\max_{\pi \in \Pi_h} J_R(\pi) \quad \text{subject to} \quad J_{C_q}(\pi) \leq d_q, \quad q = 1, \dots, Q. \quad (8)$$

Expected cost limits regulate trajectories, not individual actions. A positive violation budget permits rare violations; exactly zero expected non-negative violation implies almost-sure feasibility, but finite-sample optimisation does not certify that equality. Hard-feasible actions can still produce excessive path cost, and non-emptiness of $\mathcal{A}_h$ does not ensure feasible cumulative budgets.

2.3 PORTFOLIO INSTANTIATION

For a portfolio of N allocation entities, target weights satisfy $\mathbf{w}_t \in \Delta^N$ and asset returns $\mathbf{R}_{t+1} \in \mathbb{R}^N$ are realised after selecting $\mathbf{w}_t$. Any reserve is included as a zero-return entity within this vector. The simulator uses previous target weights, not drifted holdings:

$$T_t = \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1 \leq 2, \quad (9)$$

$$g_{t+1} = 1 + \mathbf{w}_t^\top \mathbf{R}_{t+1} - \kappa T_t, \quad \kappa = 10^{-4},$$

$$r_t = \log(\max\{10^{-6}, g_{t+1}\}),$$

$$W_{t+1} = W_t \max\{10^{-12}, g_{t+1}\}, \quad W_0 = 1. \quad (10)$$

Full- ℓ_1 turnover covers reallocation across all entities. This target-change cost is a simulator approximation: exact self-financing requires drifted holdings and fee-adjusted cash balance [9]. Slippage, tiered fees and market impact are omitted, as are the benchmark's drift-induced rebalancing fees. The growth floors avoid undefined numerical operations, not insolvency. Positive growth is ensured if $\min_{1 \leq i \leq N} (1 + R_{t+1,i}) > 2\kappa$, but Gaussian returns provide no deterministic guarantee. Without active floors, $r_t = \log(W_{t+1}/W_t)$; only the undiscounted sum telescopes to terminal log wealth.

The experimental group exposures are

$$\sum_{j \in V_1} w_{t,j} \geq 0.50, \quad V_1 = \{1, 4, 6\}; \quad \sum_{j \in V_2} w_{t,j} \geq 0.40, \quad V_2 = \{2, 4, 7\}. \quad (11)$$

Agent and benchmark start from the same feasible constrained-neutral target. The benchmark retains this target, with zero initial and ongoing target-change cost.

For agent wealth and its benchmark counterpart W_t^B , initialised at one, define

$$M_t = \max_{0 \leq u \leq t} W_u, \quad D_t = 1 - \frac{W_t}{M_t}, \quad D_t^{\max} = \max_{0 \leq u \leq t} D_u. \quad (12)$$

Benchmark quantities $M_t^B, D_t^B, D_t^{B,\max}$ are analogous. Wealth, peaks, running maximum drawdowns and previous turnover are retained in ζ_t ; the actor receives risk ratios and diagnostics rather than absolute wealth. Since peaks never fall below one, drawdown denominators remain positive.

The adaptive drawdown threshold, excess cost and cost allowance are

$$b_{t+1} = \max\{0.05, 0.90D_{t+1}^B\}, \quad c_{t+1} = \frac{[D_{t+1} - b_{t+1}]_+^2}{0.10}, \quad (13)$$

$$\alpha_{t+1} = \frac{(0.05b_{t+1})^2}{0.10}, \quad (14)$$

where $[x]_+ = \max(0, x)$. The actor observes b_t ; b_{t+1} and its cost are computed only after updating both wealth paths with the realised return. Thus the cost is non-anticipative, and the exogenous benchmark prevents the agent from directly inflating its allowance. The threshold b_t is in drawdown units, whereas α_t is in squared-cost units. The portfolio risk target is

$$J_{\text{gap}}(\pi) = \mathbb{E}_\pi \left[\frac{1}{H} \sum_{t=0}^{H-1} (c_{t+1} - \alpha_{t+1}) \right], \quad (15)$$

$$\max_{\pi \in \Pi_h} J_R(\pi) \quad \text{subject to} \quad J_{\text{gap}}(\pi) \leq 0.$$

RCPO uses undiscounted rollout-mean residuals for multiplier updates; discounted actor-critic learning is a surrogate, not an exact solution or risk certificate. The cost penalises current drawdown and can vanish after recovery, unlike running maximum drawdown. Terminal relative wealth $W_H/W_H^B - 1$ and maximum drawdown are portfolio evaluation metrics, not the general objective.

3 PROPOSED METHOD

3.1 METHOD OVERVIEW

The method implements the two constraint classes in Eq. (8) through separate mechanisms. A stochastic actor first samples dependent branch actions in a surrogate product of simplexes. CAOSD maps those branches deterministically to an allocation in $\mathcal{A}_h$, providing hard feasibility before the environment is evaluated. PPO or RCPO then updates the surrogate policy from reward and cost advantages. RCPO influences which feasible allocations are preferred over a trajectory; it is not responsible for repairing infeasible actions. Thus, every action delivered to the environment satisfies the supported group constraints independently of critic accuracy or multiplier convergence.

3.2 HARD-FEASIBLE CAOSD ACTION MAP

Consider two lower-bound constraints

$$\sum_{j \in V_1} a_j \geq \ell_1, \quad \sum_{j \in V_2} a_j \geq \ell_2, \quad (16)$$

where $V_1, V_2 \subseteq \mathcal{E}$ may overlap and $\ell_1, \ell_2 \in [0, 1]$. CAOSD constructs four padded simplex branches supported respectively on

$$K_1 = V_1 \cap V_2, \quad K_2 = V_1, \quad K_3 = V_2, \quad K_4 = \mathcal{E}. \quad (17)$$

Each branch output $y_i \in \mathbb{R}_+^N$ has support only on K_i and unit mass whenever K_i is non-empty. Let

$$q_2 = \sum_{j \in V_1 \cap V_2} y_{2,j} \quad (18)$$

denote the fraction of the second branch contributing to the second group's exposure. The implemented combination coefficients are

$$\begin{aligned} z_1 &= [\ell_1 + \ell_2 - 1]_+, & z_2 &= [\ell_1 - z_1]_+, \\ z_3 &= [\ell_2 - z_1 - z_2 q_2]_+, & z_4 &= 1 - z_1 - z_2 - z_3. \end{aligned} \quad (19)$$

The environment allocation is the weighted Minkowski sum

$$\mathbf{a}_t = \text{CAOSD}(\mathbf{y}_{1,t}, \dots, \mathbf{y}_{4,t}) = \sum_{i=1}^4 z_{i,t} \mathbf{y}_{i,t}, \quad z_{i,t} \geq 0, \quad \sum_{i=1}^4 z_{i,t} = 1. \quad (20)$$

For feasible two-group inputs, every positive-mass branch has unit mass; an empty branch can have only zero coefficient. Hence, in exact arithmetic, $\mathbf{a}_t \geq 0$ and $\mathbf{1}^\top \mathbf{a}_t = \sum_i z_{i,t} = 1$. The first group receives at least $z_1 + z_2 = \ell_1$, and the second receives at least $z_1 + z_2 q_2 + z_3 \geq \ell_2$ [1]. Numerical normalisation and feasibility diagnostics handle floating-point tolerances. This is a specialised exact map, not a solver for arbitrary polytopes or an arbitrary number of groups; methods such as PASPO address the more general case [10].

For the portfolio constraints in Eq. (11), $\ell_1 = 0.50$ and $\ell_2 = 0.40$. Hence $z_1 = 0$: the mandatory-intersection branch exists in the general construction but is inactive and receives no policy update. Branches 2–4 remain active, and the overlap still affects z_3 through q_2 .

3.3 AUTOREGRESSIVE SURROGATE POLICY

The learned stochastic action is the collection of active branch variables, not the reconstructed environment allocation. Let $\mathbf{u}_t = (\mathbf{u}_{1,t}, \dots, \mathbf{u}_{4,t})$ denote these surrogate variables and let $\mathbf{y}_{i,t}$ be their simplex-valued branch representations. The autoregressive policy factorises as

$$\pi_\theta(\mathbf{u}_t | o_t) = \prod_{i \in \mathcal{I}_{\text{active}}} \pi_{\theta,i}(\mathbf{u}_{i,t} | o_t, \mathbf{u}_{<i,t}), \quad \log \pi_\theta = \sum_{i \in \mathcal{I}_{\text{active}}} \log \pi_{\theta,i}. \quad (21)$$

This conditional structure represents dependencies induced by sequential branch allocations, especially the effect of $\mathbf{y}_2$ on q_2 and therefore on downstream available mass.

Two branch distributions are evaluated. The autoregressive Gaussian actor emits real-valued branch logits and applies a softmax within each branch. The autoregressive Dirichlet actor emits bounded positive concentration parameters and samples directly on each branch simplex. Both use a shared multilayer perceptron encoder, condition later heads on earlier realised branch weights, and apply Eq. (20) only after the complete surrogate action has been formed.

PPO likelihood ratios are computed from Eq. (21) in the sampled surrogate space. The environment-action policy in Eq. (5) is the pushforward of this law through the branch representation and CAOSD map; notation such as $J_R(\pi_\theta)$ refers to returns under that composite policy. No density is assigned directly to the final allocation, for which the deterministic CAOSD map may be many-to-one. This distinction avoids treating a reconstructed environment action as though it had been sampled from an unrelated flat distribution. The same dependency that motivates autoregression also creates a credit-assignment problem: the joint reward of $\mathbf{a}_t$ generally cannot be attributed to one branch independently of its realised mass and downstream branches.

3.4 RCPO FOR SOFT TRAJECTORY COSTS

For one cumulative constraint, a Lagrangian relaxation of Eq. (8) is

$$\mathcal{L}(\theta, \lambda) = J_R(\pi_\theta) - \lambda(J_C(\pi_\theta) - d_C), \quad \lambda \geq 0. \quad (22)$$

Here J_C is one discounted cost return and d_C its fixed budget. For the adaptive portfolio target in Eq. (15), the corresponding relaxation is $\mathcal{L}_{\text{ad}} = J_R - \lambda J_{\text{gap}}$, so the state-dependent allowance belongs inside the expectation. PPO supplies the clipped on-policy update [3]; the RCPO-inspired implementation supplies a projected dual update once per rollout [2]. In the global formulation, the actor uses the combined advantage

$$A_t^{\text{RCPO}} = A_t^R - \lambda A_t^C, \quad (23)$$

where separate critics estimate reward and cost returns. In branch-credit modes, each active branch has its own reward critic and, when counterfactual cost is used, its own cost critic. The three tested branch advantages are

$$\begin{aligned} A_{i,t}^{\text{CF-R}} &= A_{i,t}^{\Delta R} - \lambda A_t^{C,\text{global}}, \\ A_{i,t}^{\text{CF-C}} &= z_{i,t} A_{i,t}^{R,\text{standalone}} - \lambda A_{i,t}^{\Delta C}, \\ A_{i,t}^{\text{CF-RC}} &= A_{i,t}^{\Delta R} - \lambda A_{i,t}^{\Delta C}. \end{aligned} \quad (24)$$

The portfolio allowance is defined in Eq. (14). For the fixed initialisation and market-sampling protocol, its benchmark path is exogenous and the episode horizon is fixed, so the expected allowance is independent of the allocation policy. The global cost critic estimates non-negative excess cost, while counterfactual branch-cost critics estimate signed differences; neither directly estimates $c - \alpha$. However, discounted critic advantages and the undiscounted rollout-mean dual statistic are different surrogates. The combined PPO/RCPO procedure is not an exact primal-dual solver for Eq. (15), and the convergence guarantees of the original RCPO analysis are not asserted for these branch-credit and adaptive-budget variants. Once per rollout, the single global multiplier is updated using realised c_{t+1} and α_{t+1} ,

$$\lambda \leftarrow [\lambda + \eta_\lambda (\bar{c} - \bar{\alpha})]_+, \quad \eta_\lambda = \begin{cases} 0.00075, & \bar{c} > \bar{\alpha}, \\ 0.01, & \bar{c} \leq \bar{\alpha}. \end{cases} \quad (25)$$

where bars denote rollout means. Updating once per rollout prevents the number of PPO epochs or minibatches from changing the multiplier time scale. The unequal upward and downward step sizes are an implementation choice designed to let the multiplier relax faster after realised cost falls below tolerance. The multiplier remains global in every branch-credit variant because the cumulative risk in Eq. (13) is incurred by the reconstructed portfolio, not by an isolated branch. It is always updated from the non-negative actual cost, never from a signed branch counterfactual.

3.5 STANDALONE AND OPEN-LOOP COUNTERFACTUAL BRANCH CREDIT

The final allocation produces one joint reward and one joint path cost, although the actor samples several dependent branches. Standalone credit constructs a shadow trajectory in which a branch is treated as fully allocated, uses its branch-specific reward return, scales its actor signal by the realised CAOSD mass z_i , and retains the global portfolio cost. This provides a low-cost branch signal but ignores interactions through the CAOSD coefficients, transaction costs, and autoregressive successors.

Counterfactual credit instead compares the realised allocation with a matched baseline allocation, following the broader idea of holding a shared outcome context fixed while changing one component [6]. Let $\mathbf{y}_{i,t}^0$ denote the neutral allocation on branch i . The actual and branch- i counterfactual actions are

$$\begin{aligned} \mathbf{a}_t^{\text{act}} &= \text{CAOSD}(\mathbf{y}_{1,t}, \dots, \mathbf{y}_{4,t}), \\ \mathbf{a}_t^{-i} &= \text{CAOSD}(\mathbf{y}_{1,t}, \dots, \mathbf{y}_{i,t}^0, \dots, \mathbf{y}_{4,t}). \end{aligned} \quad (26)$$

The CAOSD map is recomputed rather than subtracting a padded branch from the final action; replacing a branch can change both its direct allocation and downstream masses. On the same exogenous transition, the marginal signals are

$$\Delta r_{i,t} = r_t^{\text{act}} - r_{i,t}^{\text{cf}}, \quad \Delta c_{i,t+1} = c_{t+1}^{\text{act}} - c_{i,t+1}^{\text{cf}}. \quad (27)$$

In the portfolio implementation, every counterfactual maintains its own turnover, wealth, running peak, and drawdown path, and its critic receives the corresponding counterfactual path features. All paths use the same realised market return, the accounting convention of Eq. (10), and the common exogenous benchmark budget. The cost difference is signed: neutralising a branch may increase or decrease path cost. Equation (25) nevertheless uses only the actual non-negative portfolio cost.

The construction is open-loop because later autoregressive branch actions are held at their realised values rather than resampled after replacing branch i . Because each shadow path retains its own history, cost contrasts can include accumulated consequences of earlier substitutions, not only the current branch action. The signals are path-consistent contrasts conditional on the realised other-branch sequence, not the full closed-loop effect through all downstream policy conditionals. Accordingly, the construction is a branch-credit heuristic rather than an unbiased causal estimator.

4 EXPERIMENTAL EVALUATION

4.1 PROTOCOL AND METRICS

The synthetic generator contains persistent low- and high-volatility regimes, regime-dependent relative winners, mild momentum, and regime-dependent correlations. Assets favoured in calm markets differ from the defensive assets favoured in volatile regimes, forcing the policy to infer the latent state from observations rather than receive a regime label. Training samples uniformly from eight deterministic markets, which reduces direct memorisation of one simulated path. Each 252-step episode represents one trading year and starts from the constrained-neutral allocation.

Every learned policy receives 60,000 updates with 2,048-step rollouts, minibatches of 512, hidden layers of sizes [192, 128], discount 0.99, GAE parameter 0.95, and proportional transaction cost of one basis point. Gaussian actors use three PPO epochs per rollout and Dirichlet actors use four, so environment-interaction budgets are equal but optimisation time is not. Return-best checkpoints are selected using mean excess cumulative return on ten fixed validation continuations. Final deterministic evaluation uses the same 20 disjoint future continuations for every policy; no test branch enters model selection. Holding the continuation seeds, anchor state, and transaction costs fixed isolates policy and credit-mode differences more cleanly than comparisons assembled from unrelated runs. It does not provide sampling uncertainty over policy initialisation, because the comparison uses one policy seed.

Table 1 reports mean terminal relative wealth, win rate against constrained-neutral, mean maximum drawdown (MDD), mean full- ℓ_1 turnover per step, the fraction of test branches satisfying the active drawdown-cost target, and the fraction satisfying both allocation constraints. A branch is drawdown-feasible when its episode-average c_t does not exceed its episode-average dynamic α_t ; this does not mean that every step has zero violation. Drawdown feasibility is shown only for RCPO policies trained against that target. The soft-penalty row is included to distinguish approximate constraint learning from structural feasibility and was evaluated on the same continuation paths.

Table 1: Performance on 20 shared unseen one-year continuations. All numeric entries are percentages; “–” means that the corresponding constraint was not active. “CF” denotes open-loop counterfactual credit; G and D denote Gaussian and Dirichlet actors.

Policy	Rel. wealth	Win	MDD	Turnover	DD feas.	Alloc. feas.
Constrained-neutral	0.00	–	16.70	0.00	–	100
Soft allocation penalty	5.86	80	16.99	8.65	–	0
V2.6 PPO G	4.83	75	17.99	9.16	–	100
V2.6 PPO D	5.58	70	17.41	15.47	–	100
V2.6 RCPO standalone G	4.61	70	19.14	1.91	15	100
V2.6 RCPO standalone D	8.06	90	16.73	14.27	40	100
V3.2 CF reward G	4.61	70	19.20	0.96	10	100
V3.2 CF cost G	4.97	75	18.03	7.22	30	100
V3.2 CF reward+cost G	4.13	65	18.66	4.25	10	100
V3.2 CF reward D	8.18	85	16.52	24.00	45	100
V3.2 CF cost D	6.18	85	17.00	17.14	40	100
V3.2 CF reward+cost D	6.49	75	16.85	22.89	35	100

4.2 RESULTS

The PPO comparison shows that RCPO is architecture dependent rather than uniformly beneficial. Relative to PPO, Gaussian RCPO lowers return from 4.83% to 4.61% and increases MDD from 17.99% to 19.14%, whereas Dirichlet RCPO raises return from 5.58% to 8.06% and lowers MDD from 17.41% to 16.73%. The Dirichlet RCPO control wins in 18 of 20 continuations, although only 40% meet its active drawdown-cost target. The soft allocation penalty produces competitive return but no allocation-feasible test branch. In contrast, every CAOSD policy records zero allocation violation, confirming the intended hard guarantee.

Counterfactual effects depend on the actor. For Gaussian policies, counterfactual cost raises relative wealth from 4.61% to 4.97%, lowers mean MDD from 19.14% to 18.03%, and increases drawdown-feasible branches from 15% to 30%. The improvement costs additional turnover: 7.22% rather than 1.91%. Counterfactual reward alone is essentially tied with the Gaussian control, and the joint signal weakens return and feasibility.

For Dirichlet policies, counterfactual reward attains the highest observed mean return (8.18%) and lowest MDD (16.52%) in the comparison. Its paired return difference from the V2.6 control is only +0.12 percentage points

(approximate 95% paired t -interval $[-0.87, +1.10]$), while win rate decreases from 90% to 85% and turnover rises from 14.27% to 24.00%. It is therefore not a conclusive return improvement. The corresponding Gaussian MDD difference has an approximate paired interval of $[-1.90, -0.32]$ percentage points, while its turnover increase is 5.31 percentage points. These intervals describe variation over 20 continuations conditional on one trained seed; they do not quantify training-seed uncertainty. No counterfactual variant uniformly dominates its matched standalone control.

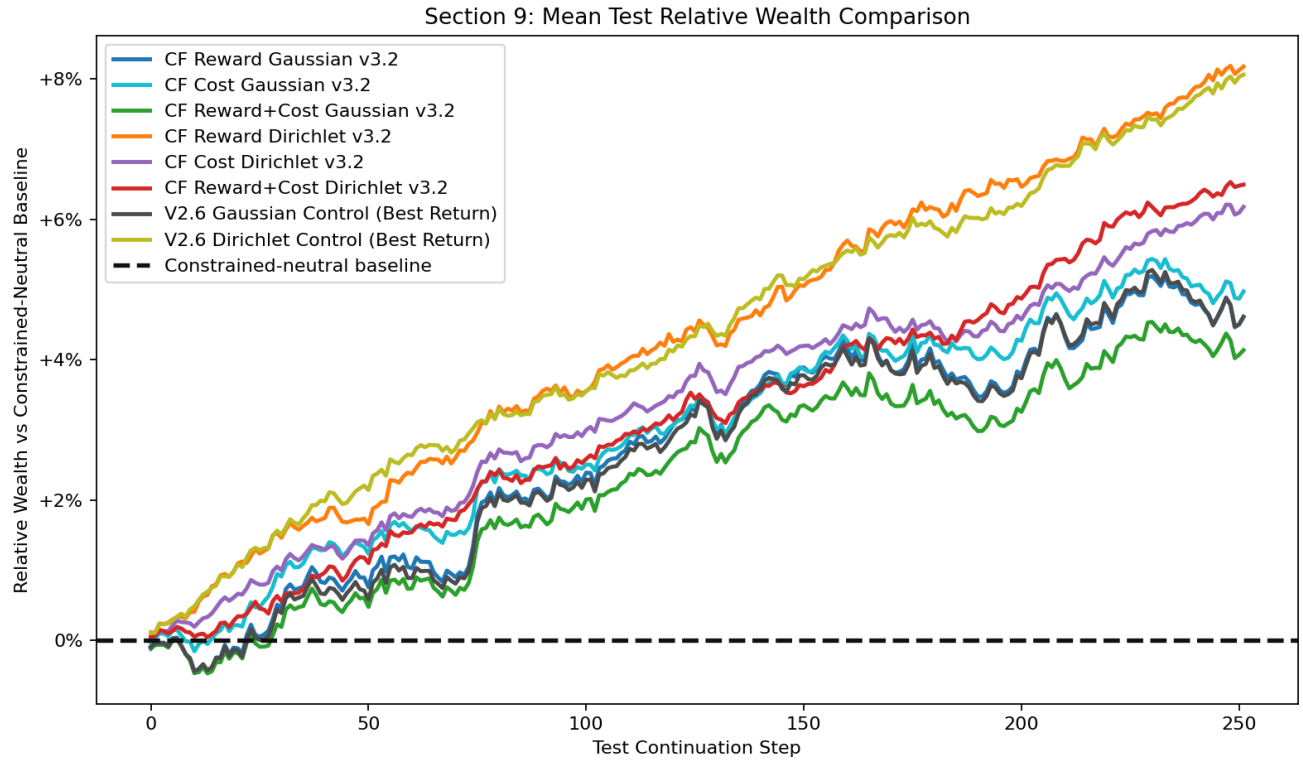


Figure 1: Mean cumulative relative wealth on 20 shared test continuations; the baseline is zero by construction. Individual branch dispersion is reported numerically rather than overplotted.

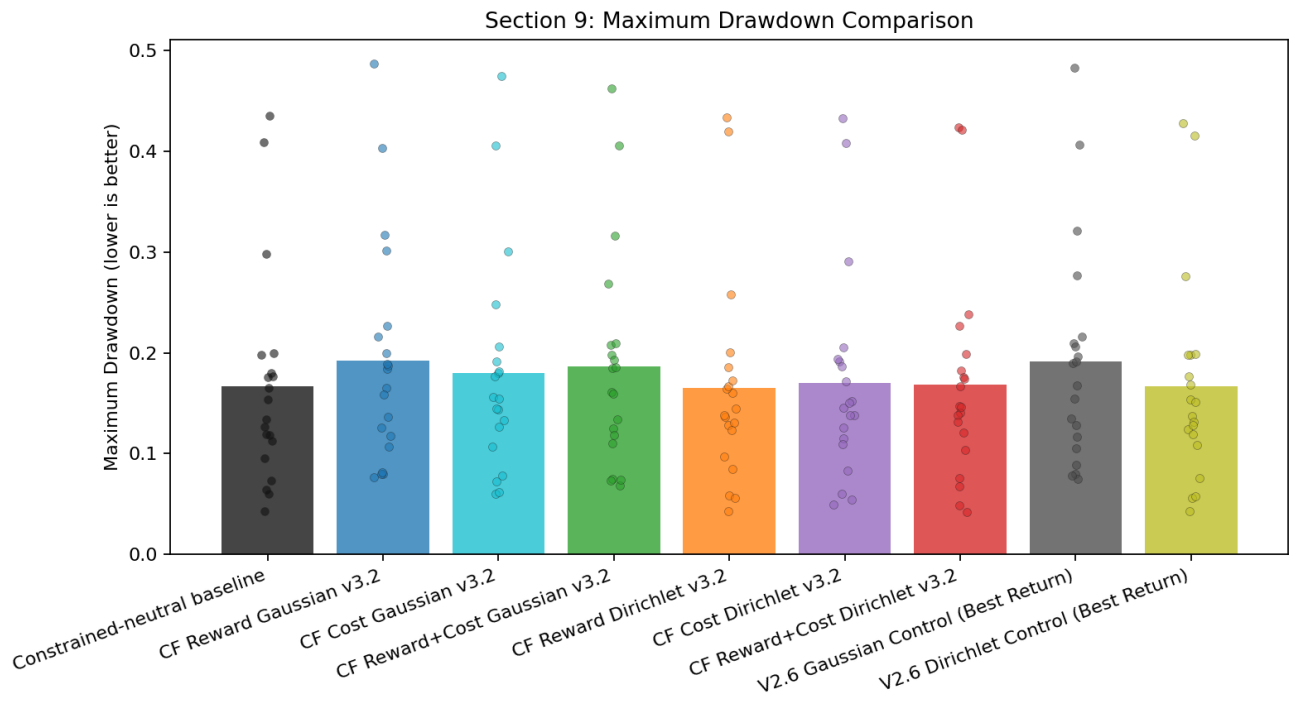


Figure 2: Mean maximum drawdown and individual continuation outcomes for V3.2 policies and matched V2.6 controls. Lower is better.

5 DISCUSSION AND LIMITATIONS

The experiments support the architectural separation underlying the method. Allocation constraints are deterministic properties of the policy output, whereas drawdown control remains distributional and empirical. A mean rollout-cost multiplier can guide behaviour but does not ensure that unseen paths satisfy the target: no V3.2 return-best policy exceeds a 45% test feasible-branch rate. A separate checkpoint rule that first maximises validation feasible-branch rate and then validation return also reaches at most 45% test feasibility; for counterfactual-cost Dirichlet it reduces test relative wealth to 0.92%. Validation-feasible selection therefore cannot be interpreted as an out-of-sample guarantee.

The counterfactual experiment provides evidence that branch-specific signals can be useful, but not that more counterfactual terms are uniformly better. Gaussian and Dirichlet actors respond differently, and the joint reward-cost construction shows no additive benefit. Increased turnover further reduces the practical significance of small gross improvements. The conclusions are bounded by one policy seed, a finite synthetic test distribution, deterministic evaluation, and a deliberately learnable market generator. Fixed validation continuations may also induce selection bias. Historical walk-forward tests, realistic execution frictions, multiple policy seeds, and branch-distribution-aware risk objectives such as chance or conditional-value-at-risk constraints are required before stronger robustness claims are justified.

The evaluation metrics should also be read jointly. Relative wealth answers whether a policy beats an equally feasible neutral allocation, MDD describes the worst peak-to-trough episode loss, and the active feasible-branch rate tests the stricter online cost target. Similar average MDDs can coexist with different feasible rates because the cost depends on when drawdown occurs relative to the benchmark, not only on its terminal maximum. Turnover supplies a further economic qualification: without a market-impact model, a high-turnover improvement is less credible than an equal-return low-turnover improvement even after the simulated proportional cost is deducted.

6 CONCLUSION

This study demonstrates a coherent hybrid design for constrained portfolio RL. CAOSD guarantees long-only simplex membership and overlapping group exposures for each action, while RCPO supplies an adaptive mechanism for benchmark-relative path risk. At equal environment-interaction budgets, counterfactual-reward Dirichlet RCPO achieves the highest observed mean relative wealth but does not conclusively improve upon its matched control; counterfactual-cost Gaussian RCPO yields the clearest risk-side improvement, at the cost of greater turnover. Joint counterfactual credit is not consistently effective, and drawdown feasibility remains unreliable. The principal result is therefore methodological rather than financial—structural constraints can be guaranteed without asking a stochastic optimiser to learn them, leaving path-dependent risk to be evaluated and improved as a separate CMDP problem.

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